Dataset 1: IMDb Top 1000
Features ”Poster_Link”, "Series_Title", "Released_Year", "Certificate", "Genre", "Runtime", "IMDB_Rating", "Meta_score", "Director", "Star1", "Star2", "Star3", "Star4", "No_of_Votes", "Gross"

This dataset had a variety of feature types, some categorical (e.g., genres and directors) and others numerical (e.g., Runtime, Year, or Rating).

There were many movies in the dataset with missing data, which were fixed using median imputation to fill the NaN values in Gross and Meta_Score. Next, I decided to convert the different strings

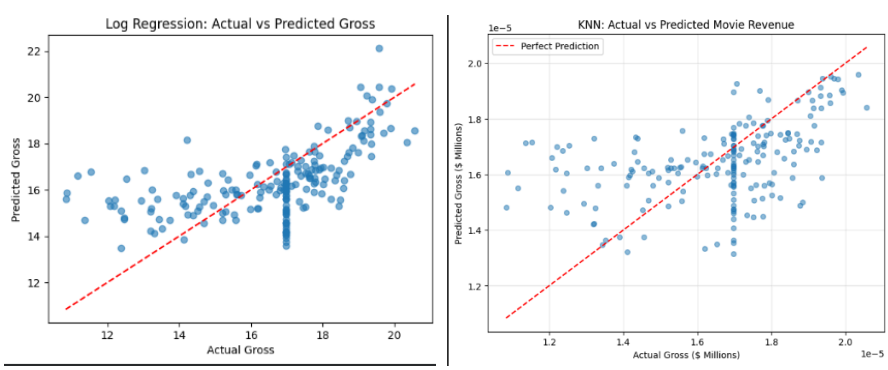
At first, I attempted to figure out if the models could predict the ratings of the movies, but I would often get really poor performance from the models. I wasn't sure why this was until I looked at the prior work another person had done using this dataset

https://www.kaggle.com/code/rmarquez1993/imdb-rating-regression-analysis

There were a few reasons why this dataset was rather difficult to work with compared to the second dataset I used.

1. The dataset was very restricted due to its only including the top 1000 movies on IMDb
2. Gross, due to Median Imputation, meant many movies would get the same gross, which can be seen in the different charts for the models

In the other work I reviewed, they actually converted the Gross using a logarithmic scale to reduce the amount it impacted outliers in the data. Making it difficult for the models to learn, the log made the model capable of learning using the Gross. This ended up improving the results significantly. The models I used were Linear regression and KNN, both of which performed well and could predict the gross income to a degree. There are very clear issues with the lower-grossing movies, due to most being mid-to-high-grossing films.



Dataset 2: Titanic Dataset
Features: “Unnamed”, “PassengerID”, “Survived”, “Sex”, “Age”, “Fare”, “Pclass_1”, “Pclass_2”, “Pclass_3”, “family_size”, “Title_1”, “Title_2”, “Title_3”, “Title_4”, “Emb_1”, “Emb_2”, “Emb_3”

I chose this dataset since I was having huge trouble with the previous IMDb dataset and wanted something simpler in comparison.

Luckily, the dataset I found was almost fully prepared for use since all the features seemed correct or usable. Only 2 columns for a bool of unnamed passengers and passengerID, which were both removed.

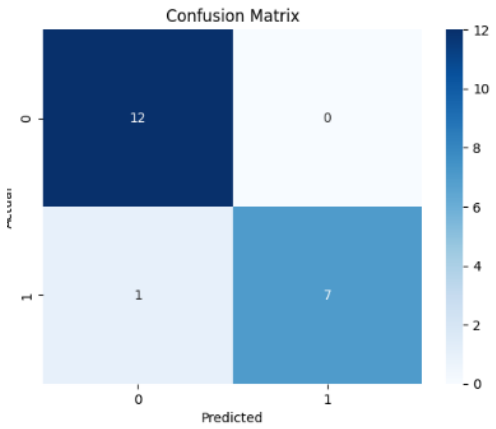
Afterwards, I created two models, one a logistic regression and the other a decision tree, and then displayed their results using the Confusion Matrix and ROC curves

The only issue with the selected dataset was the rather small amount of data it contained, only 100 rows of data; luckily, none needed to be removed, meaning that it could use all 100

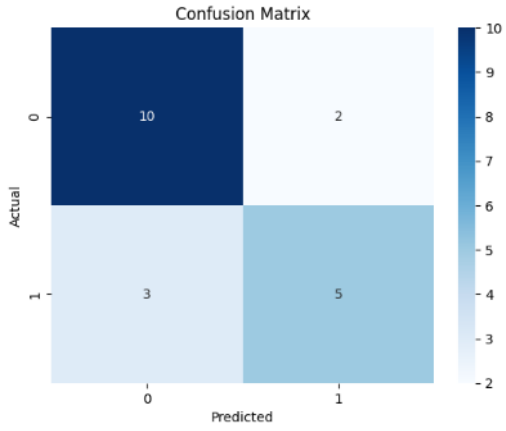
Both models performed well, with logistic regression being just slightly better than the other, which was mostly due to an overfitting issue due to the limited amount of data used to train and test the models.

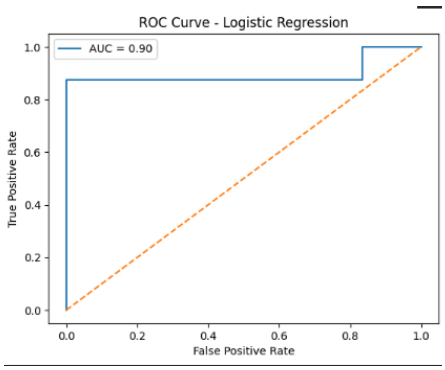
The reason it hurt the decision tree over the logistic regression is that the model needs larger amounts of data to perform better, while logistic regression could determine linear relationships faster, like gender or class, leading to higher survivial rate

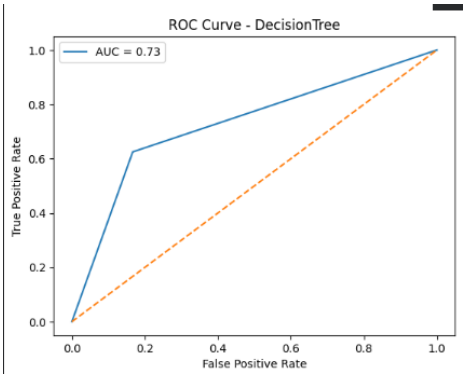
LOGISTIC REGRESSION



DECISION-TREE




ROC Curve - Logistic Regression
AUC = 0.90
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0


ROC Curve - DecisionTree
AUC = 0.73
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0